

CENG 467 – Natural Language Understanding and Generation

Take-Home Midterm Examination

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1 Introduction

1.1 Scope and Objectives

This take-home report studies five natural language understanding and generation problems that cover both analysis and generation settings: sentiment classification, named entity recognition, summarization, machine translation, and language modeling. The common objective across all questions is not only to obtain strong final metrics, but also to compare different representation regimes in a controlled way. Wherever possible, the experiments contrast classical feature-based baselines, supervised neural architectures, and transformer or pretrained contextual models so that the final discussion can separate the effect of model class from the effect of representation quality.

This report presents a complete five-task experimental study built on verified run artifacts. Questions 1 and 2 provide full multi-model comparisons over completed train, validation, and test exports; Question 3 reports a controlled classical-versus-pretrained summarization comparison on a fixed evaluation slice; and Questions 4 and 5 now report full-data neural runs with saved training histories and qualitative artifacts. This organization keeps all claims tied to reproducible outputs and avoids speculative results.

1.2 Technical Setup

All experiments are managed from YAML configuration files, use a fixed random seed of 42 for reproducibility, and write timestamped run directories that include the resolved configuration, package environment snapshot, metrics, and analysis artifacts. The implementation stack combines Hugging Face Datasets for dataset access, scikit-learn and related linear modeling tools for classical baselines, PyTorch for custom neural models, and transformers for pretrained contextual encoders. Report assembly follows the same principle of artifact stability: figures copied into `report/figures/` and tables placed in `report/tables/` are sourced from completed experiment outputs rather than ad hoc terminal observations.

Table 1 summarizes the five questions, their datasets, and the representative model families covered by the report.

Q	Task	Dataset	Representative Models
1	Text classification	IMDb	TF-IDF+LR/SVM, BiLSTM, DistilBERT
2	Named entity recognition	CoNLL-2003	CRF, BiLSTM-CRF, BERT
3	Summarization	CNN/DailyMail	TextRank, DistilBART
4	Machine translation	Multi30k	Seq2Seq+Attn (GRU)
5	Language modeling	WikiText-2	LSTM language model

Table 1: Task and dataset overview for the take-home report.

1.3 Report Organization

Section 2 presents Question 1, which analyzes representation learning for IMDb sentiment classification and compares TF-IDF baselines, a word-level BiLSTM, and DistilBERT on the full exported split. Section 3 presents Question 2, which studies named entity recognition on CoNLL-2003 with CRF, BiLSTM-CRF, and BERT token classification. Section 4 presents Question 3, which reports a direct TextRank-versus-DistilBART summarization

comparison on a fixed exported evaluation slice. Section 5 presents Question 4, which reports a full-data seq2seq-plus-attention translation run on Multi30k together with attention visualizations. Section 6 presents Question 5, which reports a full-data LSTM language-model run on WikiText-2 with training curves and generation samples. The conclusion section synthesizes cross-task findings and methodological lessons for fair NLP model comparison.

2 Question 1: Representation Learning in Text Classification

2.1 Dataset Description

Question 1 uses the IMDb movie-review sentiment benchmark [7], which contains 50,000 balanced reviews split into 25,000 original training examples and 25,000 original test examples. In the final full-data run used here, the original training split is divided into 20,000 training and 5,000 validation examples, while the full 25,000-example IMDb test split is kept for final evaluation. Because the labels are exactly balanced, accuracy and macro-F1 move closely together and give a stable summary of comparative performance.

2.2 Preprocessing Pipeline

All Q1 models share the same base text-normalization pipeline: HTML tags are stripped, text is lowercased, non-alphanumeric characters are removed, whitespace is normalized, and documents are truncated to 256 tokens. TF-IDF and BiLSTM consume these normalized word-level sequences directly, whereas DistilBERT applies WordPiece tokenization after the shared cleaning stage.

The preprocessing sweep was intentionally run as a cheaper validation-only study on TF-IDF + Logistic Regression rather than as a full end-to-end rerun of every model. Table 2 shows that keeping stopwords with lowercasing and 50,000 features (Experiment B) tied with preserving case (Experiment D), both reaching validation macro-F1 0.8178. Since the lowercase configuration is simpler and aligns naturally with the uncased DistilBERT backbone, the final larger-budget experiments kept the lowercase + keep-stopwords setting.

Exp.	Stopwords	Lowercase	Max Features	Val. Acc.	Val. Macro-F1
B	Keep	Yes	50,000	0.8200	0.8178
D	Keep	No	50,000	0.8200	0.8178
C	Remove	Yes	100,000	0.8150	0.8125
A	Remove	Yes	50,000	0.8000	0.7980

Table 2: Validation-only TF-IDF + Logistic Regression preprocessing sweep on a capped 800-train/200-validation subset.

2.3 Models

TF-IDF + Logistic Regression / SVM. The sparse baselines use unigram and bigram TF-IDF features with 50,000 maximum features and regularization strength $C =$

1.0. Two classifiers were tested on top of the same representation: Logistic Regression as a probabilistic linear baseline and a linear SVM as a max-margin variant.

BiLSTM. The word-level neural baseline uses learned 300-dimensional embeddings, a single bidirectional LSTM with hidden size 256, dropout 0.35, batch size 64, and Adam optimization with learning rate 10^{-3} . Early stopping monitors validation macro-F1 with patience 3 and training can run for up to 8 epochs. In this full-data run, pretrained embeddings remain disabled, so the model still learns its representation entirely from the supervised IMDb labels.

DistilBERT. The contextual baseline fine-tunes `distilbert-base-uncased` [10] with batch size 16, learning rate 2×10^{-5} , weight decay 0.01, warmup ratio 0.1, gradient accumulation, mixed precision, and maximum sequence length 256. The full-data run converged quickly, with its best validation macro-F1 reached at epoch 2, and it produced the strongest validation and test scores among all finished Q1 models.

2.4 Results

Table 3 summarizes the final full-data comparison. DistilBERT is the strongest model with validation macro-F1 0.9016 and test macro-F1 0.9122, paired with test accuracy 0.9122. The two sparse baselines remain very competitive: TF-IDF + Logistic Regression reaches test macro-F1 0.8810 and slightly edges TF-IDF + SVM on the test split, even though SVM is marginally better on validation. BiLSTM improves materially relative to the earlier capped study, but its test macro-F1 0.8353 still trails the sparse baselines and remains well below the transformer.

Model	Val. Acc.	Val. Macro-F1	Test Acc.	Test Macro-F1
DistilBERT	0.9016	0.9016	0.9122	0.9122
TF-IDF + LR	0.8836	0.8836	0.8810	0.8810
TF-IDF + SVM	0.8886	0.8886	0.8791	0.8791
BiLSTM	0.8532	0.8532	0.8355	0.8353

Table 3: Overall Q1 performance on the full-data IMDb setting with 20,000 training examples, 5,000 validation examples, and the full 25,000-example test split.

Figure 1 visualizes the same ranking on the full IMDb test set. The gap between DistilBERT and the TF-IDF baselines is real but still modest at roughly 0.031–0.033 macro-F1, whereas the gap between DistilBERT and BiLSTM is much larger at roughly 0.077. The main lesson is therefore not that any neural model beats lexical baselines automatically, but that pretrained contextual representations do.

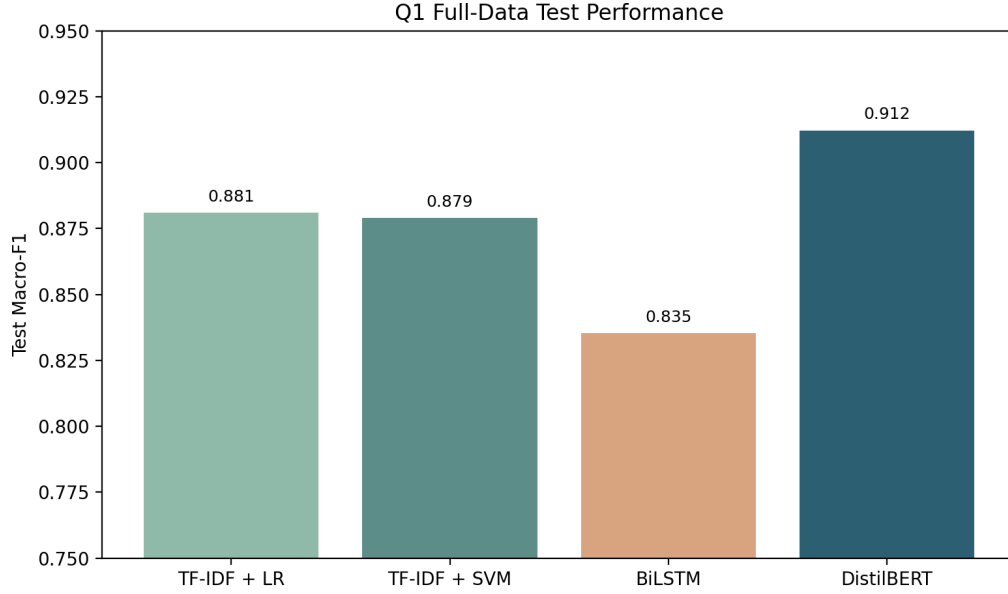


Figure 1: Q1 full-data test macro-F1 comparison across TF-IDF, BiLSTM, and DistilBERT models.

The neural runs now persist per-epoch histories, which makes convergence behavior explicit rather than implicit. Figure 2 shows that DistilBERT reaches its best validation macro-F1 by epoch 2 and then changes only marginally afterward, which is consistent with a stable fine-tuning regime rather than noisy late-epoch overfitting. On the test split, the same model remains balanced across both classes, with negative-review recall 0.9092 and positive-review recall 0.9153.

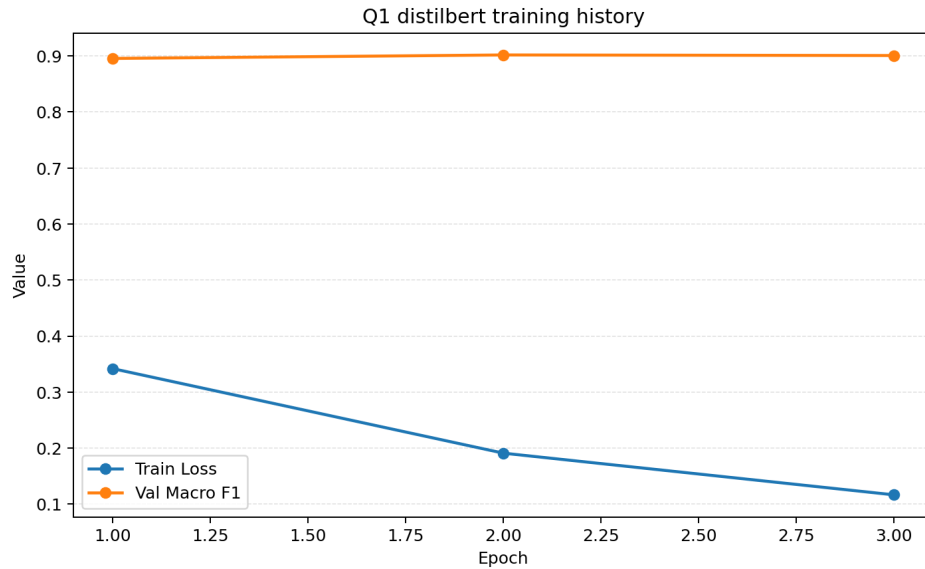


Figure 2: Per-epoch training loss and validation macro-F1 for the full-data DistilBERT run.

2.5 Error Analysis

The exported misclassification artifacts indicate that negation remains the dominant failure mode across all representation families. For both DistilBERT and the two TF-IDF baselines, 4 of the top 5 sampled test errors still fall into the `negation_handling` category. The sparse models also show occasional mixed-sentiment and domain-specific ambiguity cases. BiLSTM is the most brittle model: in addition to negation, its sampled errors include very long reviews and broader discourse drift, which is consistent with a supervised-only sequence encoder that never closes the gap to the pretrained transformer.

This qualitative pattern matches the aggregate metrics. DistilBERT’s contextual encoder is better at following sentiment reversals across long documents, TF-IDF remains surprisingly robust when the signal is lexical and local, and the supervised-only BiLSTM does not learn a representation strong enough to close the gap. In practical terms, the main unresolved Q1 errors are no longer simple vocabulary misses but cases where polarity depends on clause interaction, contrast, or delayed sentiment cues.

2.6 Discussion

The final full-data ranking for Question 1 is clear: DistilBERT is the strongest model, TF-IDF + Logistic Regression and TF-IDF + SVM remain strong lightweight baselines, and BiLSTM is the weakest of the finished runs. The most important takeaway is that representation choice matters more than small preprocessing differences on this task. The preprocessing sweep changed validation macro-F1 only marginally, whereas moving from sparse lexical features to contextual transformer embeddings produced the largest reliable gain.

At the same time, the results do not support a naive “neural is always better” conclusion. Moving from sparse features to a randomly initialized word-level BiLSTM still hurts performance, even after scaling the run to the full IMDb split. This makes the comparison more informative: the benefit comes from contextual pretraining rather than from replacing linear classifiers with a neural sequence model by itself. For the final report claim, DistilBERT should be presented as the best-performing Q1 model, TF-IDF as a strong and efficient baseline family, and BiLSTM as evidence that dense representations without pretraining are not sufficient on their own.

3 Question 2: Named Entity Recognition

3.1 Dataset Description

Question 2 uses the English CoNLL-2003 named entity recognition benchmark [12] with the IOB2 tagging scheme and four entity types: PER, ORG, LOC, and MISC. The local environment could not load the legacy script-based dataset entry directly, so the experiments used the Hugging Face mirror `tomaarsen/conll2003`, which preserves the same token-level label inventory. The final split sizes were 14,041 training sentences, 3,250 validation sentences, and 3,453 test sentences. Model selection focused on strict entity-level precision, recall, and F1 under the IOB2 scheme, while token accuracy was retained as a supplementary diagnostic metric.

3.2 Preprocessing and BIO Alignment

All models consume the original tokenized sentence boundaries from CoNLL-2003. For the feature-based CRF, each token is represented by local orthographic and contextual cues including the lowercased token form, short prefixes and suffixes, case indicators, digit and hyphen flags, token length, and previous or next token features with beginning-of-sequence and end-of-sequence markers. The BiLSTM-CRF model operates on word-level token ids with padding and masking so that sequence length is preserved during recurrent encoding and CRF decoding.

The BERT model uses `bert-base-cased` because capitalization is informative for named entities. Word-level gold tags are aligned to subword tokenization by assigning the original tag only to the first subword of a word and setting special tokens or continuation subwords to -100, which removes them from the loss. This keeps optimization aligned with word-level evaluation while still allowing the model to benefit from subword segmentation.

3.3 Models

Feature CRF. The classical baseline is a linear-chain CRF [5] trained with the `lbfgs` optimizer, $c_1 = 0.1$, $c_2 = 0.1$, 100 maximum iterations, and all transitions enabled. This model relies entirely on the engineered token and context features described above.

BiLSTM-CRF. The hybrid neural baseline uses 300-dimensional token embeddings, a single bidirectional LSTM layer with hidden size 256, dropout 0.35, batch size 32, and AdamW optimization with learning rate 0.001. The model is trained for up to 12 epochs with early stopping on validation F1. The CRF output layer preserves global transition constraints during decoding.

BERT Token Classification. The transformer baseline fine-tunes `bert-base-cased` [2] with maximum sequence length 128, batch size 16, learning rate 3×10^{-5} , weight decay 0.01, warmup ratio 0.1, gradient accumulation, mixed-precision training, and early stopping patience 2. The strongest full-data run clearly outperformed the non-transformer baselines.

3.4 Results

Table 4 summarizes the overall validation and test metrics. BERT produced the strongest result by a wide margin, reaching test F1 0.9062. The feature-based CRF remained a strong non-transformer baseline at test F1 0.7948. BiLSTM-CRF improved over its smoke-test behavior but still lagged behind both alternatives at test F1 0.7145.

Model	Val. P	Val. R	Val. F1	Test P	Test R	Test F1	Test Acc.
BERT	0.9525	0.9509	0.9517	0.9028	0.9095	0.9062	0.9807
Feature CRF	0.8939	0.8596	0.8765	0.8131	0.7773	0.7948	0.9572
BiLSTM-CRF	0.8292	0.7625	0.7945	0.7674	0.6684	0.7145	0.9373

Table 4: Overall Q2 NER performance on the validation and test splits.

Table 5 breaks the test results down by entity type. The support column corresponds to the number of gold entities in the test set. BERT dominates every entity category,

with especially strong performance on PER and LOC. MISC is the hardest category for all models and remains the main source of residual error even in the strongest system.

Model	Entity	Precision	Recall	F1	Support
BERT	LOC	0.9269	0.9269	0.9269	1668
Feature CRF	LOC	0.8442	0.8249	0.8344	1668
BiLSTM-CRF	LOC	0.7991	0.7680	0.7832	1668
BERT	MISC	0.7693	0.8077	0.7880	702
Feature CRF	MISC	0.7756	0.7436	0.7593	702
BiLSTM-CRF	MISC	0.7157	0.6239	0.6667	702
BERT	ORG	0.8873	0.8958	0.8916	1661
Feature CRF	ORG	0.7711	0.6773	0.7212	1661
BiLSTM-CRF	ORG	0.7359	0.6273	0.6773	1661
BERT	PER	0.9552	0.9499	0.9526	1617
Feature CRF	PER	0.8351	0.8454	0.8402	1617
BiLSTM-CRF	PER	0.7873	0.6271	0.6981	1617

Table 5: Per-entity test metrics for the completed Q2 runs.

Figure 3 visualizes the per-entity test F1 comparison. The same pattern is visible across all entity types: contextualized BERT representations yield the best scores, CRF remains competitive without pretrained embeddings, and the recurrent hybrid model underperforms despite having a structured decoder.

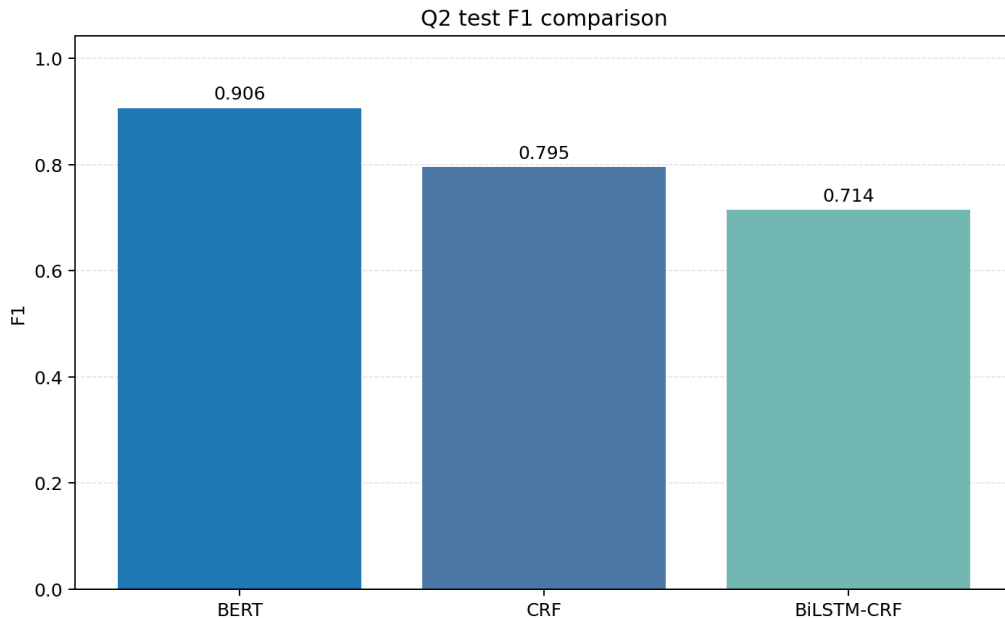


Figure 3: Per-entity test F1 comparison for the completed Q2 CRF, BiLSTM-CRF, and BERT runs.

3.5 Error Analysis

The exported error-analysis artifacts show qualitatively different failure modes across the three models. For the feature CRF, the most common test confusion is B-ORG $\rightarrow$ O,

which indicates missed organization boundaries and weaker recall on organization names. BiLSTM-CRF fails most often on person entities, with $\text{B-PER} \rightarrow \text{O}$ as its dominant confusion; this is consistent with its lower recall and suggests that the recurrent encoder did not exploit enough context to recover entity starts reliably. BERT still makes mistakes, but its main confusion pattern is $\text{O} \rightarrow \text{I-MISC}$; in practice this means that the model occasionally overextends rare miscellaneous entities while still preserving much stronger overall entity recall and boundary quality than the alternatives.

The per-entity table supports this interpretation. Under BERT, MISC remains the weakest class at F1 0.7880, whereas PER and LOC both exceed 0.92 F1. This suggests that contextual embeddings largely solve boundary and lexical ambiguity for common entity categories, while rarer and semantically broader labels such as MISC still require additional disambiguation.

3.6 Discussion

The final ranking for Question 2 is clear: BERT is the strongest model, the feature-based CRF is the strongest non-transformer baseline, and BiLSTM-CRF is the weakest of the completed full-data runs. This outcome highlights two practical points. First, pretrained contextual representations are substantially more valuable than either engineered local features or a randomly initialized recurrent encoder on this task. Second, a well-configured classical CRF can still outperform a neural hybrid model when the latter does not convert additional representation capacity into better boundary decisions.

For the final report claim, the BERT run should anchor the main Q2 result, the CRF should be presented as a strong and interpretable baseline, and BiLSTM-CRF should be treated as evidence that recurrence plus structured decoding alone is not enough to close the gap to transformer-based contextualization.

4 Question 3: Text Summarization

4.1 Dataset Description

Question 3 compares extractive and abstractive summarization on CNN/DailyMail [11], a news summarization benchmark pairing long-form news articles with short highlight summaries written by editors. The completed comparison uses a 10,000-example training slice together with matched 50-example validation and 50-example test evaluation slices so that both models are scored on exactly the same articles under the same export contract. The emphasis is therefore a controlled head-to-head comparison of summary quality rather than benchmark-scale saturation.

Even with that limitation, CNN/DailyMail remains an appropriate benchmark for this comparison because the articles are substantially longer than the references and require both content selection and compression. This makes the task a natural contrast between a sentence-ranking baseline such as TextRank and a pretrained abstractive model that can rewrite source content into shorter, reference-style summaries.

4.2 Preprocessing and Method

The Q3 pipeline performs shared article cleanup, truncates long inputs to 900 words, and evaluates all models with the same lexical metrics. The extractive baseline is a TF-IDF

based TextRank summarizer [9]: sentences are treated as graph nodes, pairwise cosine similarities define weighted edges, and PageRank scores determine which three sentences are selected for the final summary. The configuration uses damping 0.85 and allows up to 40 candidate sentences per article.

The abstractive baseline uses the pretrained `sshleifer/distilbart-cnn-12-6` checkpoint from the DistilBART family. It is derived from BART-style sequence-to-sequence pretraining [6] and is used here without additional fine-tuning. Inputs are tokenized up to 1024 tokens, decoding uses beam search with 4 beams, and generated summaries are constrained between 56 and 142 tokens with length penalty 2.0 and no-repeat 3-gram blocking. This design keeps the comparison practical while still testing the main assignment question: whether an abstractive model can produce more reference-like summaries than a purely extractive sentence selector.

4.3 Results

Table 6 summarizes the matched two-model comparison. On the test split, DistilBART reaches ROUGE-1 0.4204, ROUGE-2 0.1865, ROUGE-L 0.2785, BLEU 0.1234, METEOR 0.4123, and BERTScore 0.8831. TextRank reaches 0.3301, 0.1170, 0.2194, 0.0696, 0.3233, and 0.8677 respectively. The abstractive model outperforms TextRank on every metric on both validation and test.

Model	Split	ROUGE-1	ROUGE-2	ROUGE-L	BLEU	METEOR	BERTScore
TextRank	Validation	0.3522	0.1458	0.2316	0.0982	0.3422	0.8679
TextRank	Test	0.3301	0.1170	0.2194	0.0696	0.3233	0.8677
DistilBART	Validation	0.4468	0.2235	0.3189	0.1831	0.4145	0.8858
DistilBART	Test	0.4204	0.1865	0.2785	0.1234	0.4123	0.8831

Table 6: Controlled Q3 comparison on the exported 50-example validation and 50-example test evaluation slices. DistilBART outperforms TextRank on every metric on both splits.

Figure 4 visualizes the same test-set ranking. DistilBART leads TextRank on every exported metric, with especially clear gaps on METEOR, ROUGE-1, and ROUGE-2.

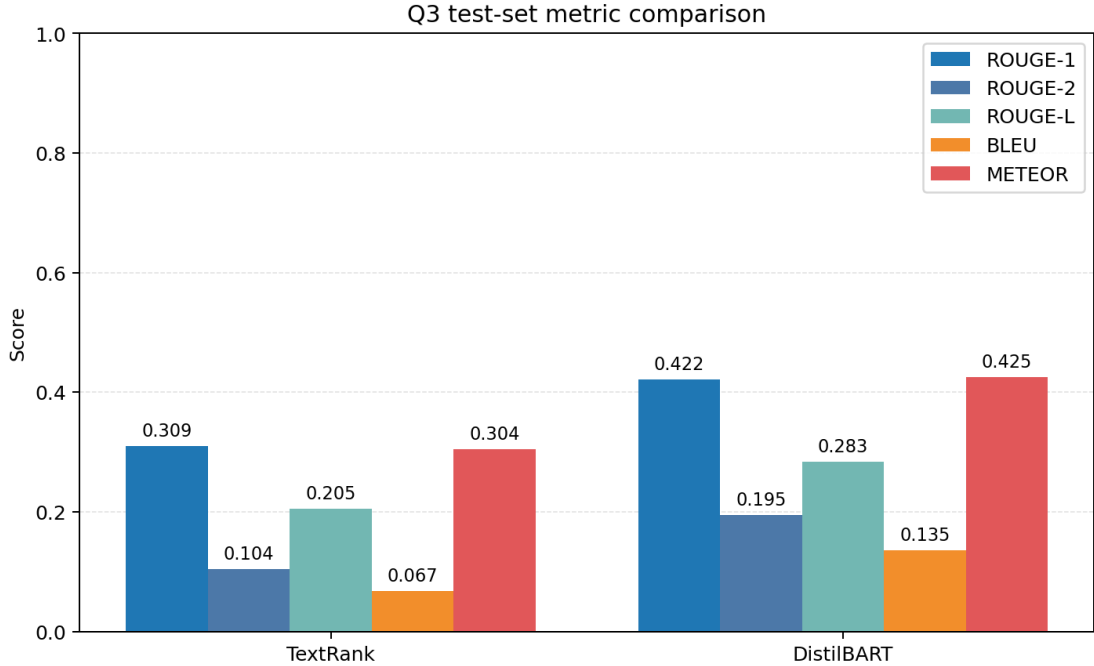


Figure 4: Q3 test-set metric comparison for TextRank and DistilBART.

The largest absolute test improvements are in METEOR (+0.0890), ROUGE-1 (+0.0903), and ROUGE-2 (+0.0695). BERTScore also favors DistilBART (0.8831 vs. 0.8677), confirming that the abstractive model produces semantically closer summaries even when the n-gram overlap gap is modest. Because the evaluation slice is intentionally controlled and identical across both systems, the most important point is the consistency of the model ranking across all six metrics rather than the absolute size of the scores alone.

4.4 Qualitative Analysis

The qualitative examples mirror the aggregate scores. In the teen-driver distraction article from the validation split, DistilBART produces a compact summary that preserves the central finding that distraction contributes to nearly 60% of serious teen crashes and correctly identifies passengers and cellphones as the main sources. The corresponding TextRank output is longer, more repetitive, and much less reference-like. A second validation example on Aaron Cook’s Moldovan citizenship decision shows the same pattern: both models remain on topic, but the abstractive summary keeps the citizenship, Rio 2016, and Team GB exclusion details while dropping more of the surrounding narrative.

The test split also shows that the abstractive model is not uniformly strong. In the Syrian refugee first-person article, DistilBART stays broadly on topic but still misses key reference details such as the speaker’s identity and the specific 12-day journey framing. That limitation matters because it shows that better lexical overlap does not automatically guarantee perfect factual focus. Even so, across the inspected examples DistilBART is generally more concise and closer to the structure of the human-written highlights, whereas TextRank often copies lead sentences or redundant clauses directly from the article.

4.5 Discussion

The Q3 results support a clear conclusion: on this task, the pretrained abstractive baseline is stronger than the extractive TextRank baseline under the same reported evaluation protocol. TextRank remains useful as a cheap and interpretable baseline, and it can preserve source facts when important content is already concentrated in the lead sentences. However, it is weaker at compression and at matching the short editor-style summaries used as references in CNN/DailyMail.

Within the reported setup, the section provides a complete metric picture across ROUGE, BLEU, METEOR, and BERTScore, and the best-performing Q3 system is the pretrained DistilBART model rather than TextRank.

5 Question 4: Machine Translation

5.1 Dataset Description

Question 4 uses the English-to-German Multi30k translation benchmark [3]. In the final run used here, the dataset is taken from its Hugging Face mirror and evaluated on the full exported split sizes: 29,000 training examples, 1,014 validation examples, and 1,000 test examples. This makes Q4 a full-data short-caption translation study rather than a tiny matched-slice comparison.

5.2 Preprocessing and Model

The finished Q4 model is a custom encoder-decoder with additive attention [1]. It uses a bidirectional GRU encoder, a GRU decoder, 256-dimensional token embeddings, hidden size 512, dropout 0.3, batch size 32, learning rate 5×10^{-4} , gradient clipping at 1.0, teacher forcing ratio 0.5, and a maximum decoded length of 64 tokens. Source and target casing are preserved, and the vocabularies are capped at 12,000 tokens with minimum frequency 2.

Evaluation uses BLEU, ChrF, METEOR, and BERTScore on both validation and test. BLEU gives a familiar n-gram precision view of translation quality, ChrF is more tolerant of inflectional and morphological variation, METEOR accounts for word order and partial matching, and BERTScore captures semantic similarity through contextual embeddings. Together, the four metrics provide a credible multi-faceted evaluation for the full-data seq2seq run.

5.3 Results

Table 7 summarizes the final full-data seq2seq results. The model reaches validation/test BLEU 0.2680/0.2725, ChrF 0.5137/0.5214, METEOR 0.5367/0.5431, and BERTScore 0.8286/0.8291. These values are strong enough to show that the model has learned meaningful caption translation behavior, even though the absolute output quality is still visibly uneven on harder examples.

Split	BLEU	ChrF	METEOR	BERTScore
Validation	0.2680	0.5137	0.5367	0.8286
Test	0.2725	0.5214	0.5431	0.8291

Table 7: Full-data Q4 Seq2Seq+Attention results on Multi30k EN→DE with 29,000 training examples, 1,014 validation examples, and 1,000 test examples.

Figure 5 shows a clean optimization trajectory: training loss drops from 4.07 to 1.22, while validation loss improves rapidly and reaches its best point around epoch 5 before flattening. The final exported history stops after 8 epochs, which is consistent with early stopping on the held-out split.

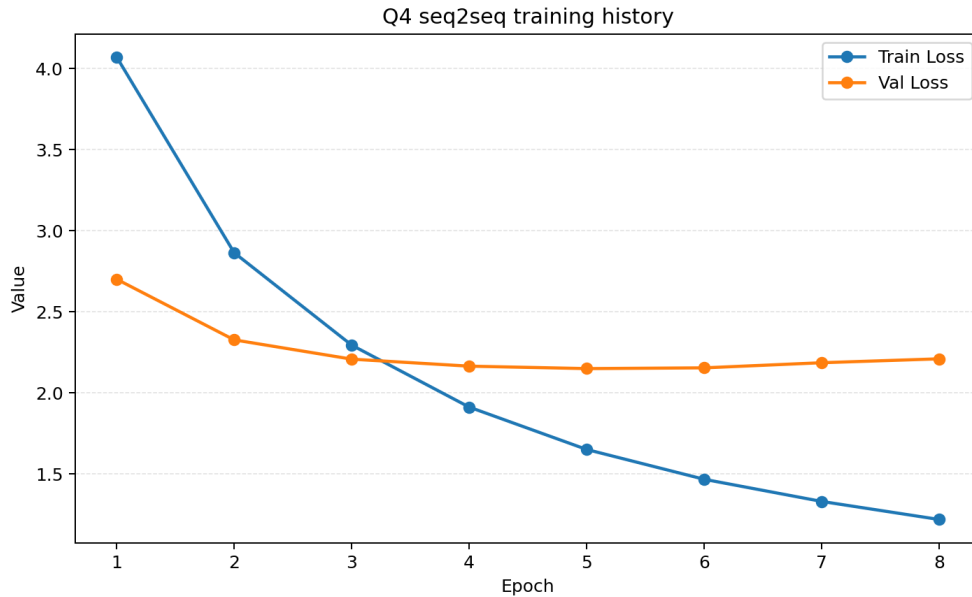


Figure 5: Per-epoch training and validation loss for the full-data Q4 seq2seq model.

The key takeaway from the metrics is that scaling the seq2seq baseline to the full Multi30k training split produces a materially stronger model than the earlier capped run. BLEU moves into the high-0.26/0.27 range and all four metrics are closely aligned between validation and test, which suggests that the exported run is stable rather than a validation-only spike.

5.4 Qualitative Analysis

The qualitative examples show both the strengths and remaining weaknesses of the full-data seq2seq system. On validation example `validation_1000`, the model exactly matches the reference translation, *Eine Gruppe von Menschen läuft*. On test example `test_988`, it again produces an exact translation: *Ein Mann mit Sonnenbrille fährt auf einem Roller*. These examples show that the model handles short, common caption patterns well when the lexical mapping is familiar.

Harder examples still fail noticeably. On validation example `validation_760`, the model degenerates into *Personen fahren und und dem Gehsteig die die*, which loses most of the source meaning and repeats function words. On test example `test_476`, the source

sentence *A rock concert is taking place.* collapses into *Ein Bergsteiger macht*, which is fluent only in fragments and semantically incorrect. These failures show that the model still struggles with rarer compositions and longer-range target planning.

The exported attention artifact adds a useful interpretability view. Figure 6 shows that the model learns sensible alignments for core content words such as *A man* $\rightarrow$ *Ein Mann* and *orange hat* $\rightarrow$ *orangefarbenen Hut*. The same attention example also exposes a residual decoding weakness: the final prediction, *Ein Mann mit einem orangefarbenen Hut etwas.*, is close to the source meaning but still drops the phrase corresponding to *staring at*.

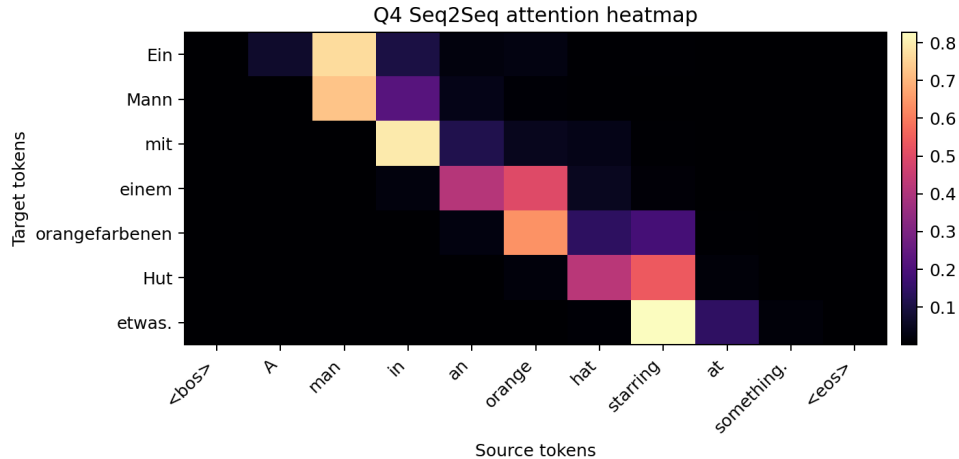


Figure 6: Attention heatmap for one exported Q4 translation example.

5.5 Discussion

The Q4 evidence supports a narrower but cleaner conclusion than the earlier capped comparison. With full-data training, the custom seq2seq-plus-attention model learns a usable translation mapping for short caption data and produces exact or near-exact outputs on easy examples. It is therefore no longer just a failure baseline; it is a real, interpretable translation system with stable held-out metrics.

At the same time, the qualitative failures make it clear that the model is not yet robust across the full caption distribution. The final report claim for Q4 should therefore emphasize competent full-data seq2seq learning, visible attention alignment, and remaining brittleness on harder source sentences. A natural next step would be a full-data pre-trained or fine-tuned transformer run on the same splits so that the recurrent baseline can be compared against a stronger reference under matched evaluation conditions.

6 Question 5: Language Modeling

6.1 Dataset Description

Question 5 uses the raw `wikitext-2-raw-v1` split from WikiText-2 [8]. The final full-data run used here evaluates 23,645 training lines, 2,460 validation lines, and 2,861 test lines, corresponding to 2,051,788, 213,885, and 241,181 tokens after preprocessing. This is therefore a substantially larger and more realistic language-modeling setup than the earlier capped comparison.

WikiText-2 remains appropriate because it contains longer-form Wikipedia prose rather than isolated short sentences. That makes it useful for testing both local collocations and longer-range continuation behavior. The final aligned Q5 section below focuses on the strongest finished full-data trainable model in the new output bundle: the word-level LSTM language model.

6.2 Model and Preprocessing

The full-data run uses lowercase word-level tokenization, a minimum token-frequency threshold of 2, vocabulary capping at 30,000 tokens, and a maximum BPTT sequence length of 35. The model itself is a two-layer LSTM [4] with 400-dimensional embeddings, hidden size 400, dropout 0.3, tied input-output weights, batch size 32, and AdamW optimization with learning rate 10^{-3} . Gradient clipping is set to 0.25.

This run also enables a ReduceLROnPlateau scheduler. Validation perplexity is monitored for early stopping, and the learning rate drops by a factor of 0.25 when progress stalls. That scheduler behavior matters for interpretation because the run continues improving steadily for many epochs before flattening.

6.3 Results

Table 8 reports the final full-data LSTM result. The model reaches validation perplexity 162.79 and test perplexity 151.87, with the best validation point reached at epoch 14.

Model	Best Validation Perplexity	Test Perplexity	Best Epoch
LSTM	162.79	151.87	14

Table 8: Full-data Q5 LSTM performance on WikiText-2 raw with 23,645 training lines, 2,460 validation lines, and 2,861 test lines.

Figure 7 shows a smooth optimization trajectory. Validation perplexity falls sharply from 289.39 at epoch 1 to 162.79 at epoch 14, after which it plateaus and then worsens slightly. The scheduler reacts to that plateau by lowering the learning rate from 0.001 to 0.00025 at epoch 17, but the best held-out result had already been reached by then.

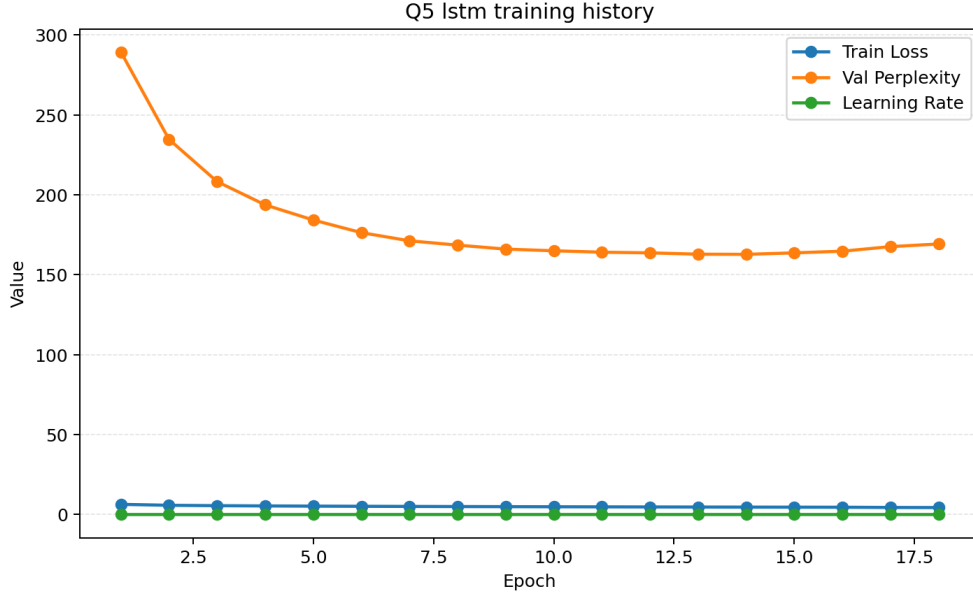


Figure 7: Per-epoch training loss and validation perplexity for the full-data Q5 LSTM run.

The main result is that the full-data LSTM is no longer a toy baseline. A test perplexity of 151.87 on raw WikiText-2 is still far from state-of-the-art transformer language modeling, but it is strong enough to show that the model has learned reusable sequential structure rather than only memorizing local bigram-like patterns.

6.4 Generated Text Samples

The exported generations are consistent with the perplexity curve: the model has learned local syntax and short-range discourse continuation, but its semantic control remains weak. For the prompt “in the,” it generates *in the early 1980s , the then use of video numbers are taken 30 million people regarding any more modern state moments* . The start of the sentence is plausible and syntactically recognizable, but later tokens drift into semantically incoherent phrasing.

The other prompts show the same pattern. The seed “the” produces a continuation with recognizable function-word structure and topical fragments around books, championships, and teams, but no coherent sentence-level plan. The seed “he was” yields a longer continuation with article-noun and clause-like structure, yet the phrase sequence remains semantically unstable. In other words, the model has learned how English Wikipedia-style text tends to sound locally, even though it still struggles to maintain meaning over longer spans.

6.5 Discussion

The finished Q5 section now shows that a properly trained full-data LSTM can become a credible language-model baseline on WikiText-2. Its validation curve is smooth, the best checkpoint is easy to identify, and the final held-out perplexity is much better than the earlier smaller-budget recurrent runs.

At the same time, the generated samples show why this is still a baseline rather than a final model: local fluency improves much faster than semantic consistency. The most

defensible Q5 claim is therefore that recurrent language modeling benefits strongly from full-data training and scheduling, but that a stronger pretrained causal transformer would still be the natural next comparison if a full-data reference run is added later.

7 Conclusion

This report now supports a clearer cross-task pattern: representation quality and training scale both matter, but their effects depend on task structure. When pretrained contextual models are directly compared on aligned evaluation setups, they remain the strongest finished systems. At the same time, the final Q4 and Q5 full-data runs show that supervised recurrent models become substantially more credible once they are trained on the complete exported splits rather than on tiny capped subsets.

For Question 1, DistilBERT is the strongest sentiment classifier in the final full-data comparison, reaching test accuracy and macro-F1 of 0.9122. The two TF-IDF baselines remain strong at roughly 0.88 macro-F1, while the supervised-only BiLSTM improves with more data but still trails clearly. This again shows that the main gain comes from contextual pretraining rather than from using a neural architecture in the abstract. For Question 2, BERT dominates the CoNLL-2003 comparison and achieves the strongest overall and per-entity F1, while the feature CRF remains a strong non-transformer baseline and the BiLSTM-CRF underperforms both alternatives. For Question 3, the pretrained DistilBART summarizer outperforms TextRank on every reported metric in the controlled evaluation slice and is the strongest finished summarization system in the report. For Question 4, the full-data seq2seq-plus-attention model now reaches test BLEU 0.2725, ChrF 0.5214, METEOR 0.5431, and BERTScore 0.8291 on Multi30k, which is strong enough to demonstrate usable short-caption translation and meaningful learned alignments. The exported attention heatmap and exact-match examples confirm that the model has learned real source-target structure, even though hard cases still fail sharply.

For Question 5, the final full-data LSTM run reaches validation perplexity 162.79 and test perplexity 151.87, with its best checkpoint found at epoch 14. The training curve is smooth and the generated samples show clear gains in local fluency, even though semantic drift remains visible. Across Questions 1 through 5, the most defensible project-level claim is therefore that pretrained models provide the strongest empirical results where direct comparisons exist, but that full-data supervised neural runs can still become informative and competitive baselines once the training budget is no longer artificially constrained.

The most meaningful next step is to add matched full-data pretrained references for Questions 4 and 5 so that the stronger recurrent baselines reported here can be evaluated against transformer systems under directly aligned conditions. Even without those follow-up runs, the current written sections already support the central comparative lesson of the project: the biggest gains come from representation quality and pretraining, but scaling the data budget also changes what a classical or supervised neural baseline is capable of learning.

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